# Problem 2: Uncertainty Analysis of the Zinc Concentration Estimation

Detailed analytical solution by the GUM method

## 1 Problem statement

A high-purity zinc sample is weighed and dissolved in a specific volume of liquid solvent. The concentration of zinc in the solution is modelled as

$$c_{Zn} = \frac{1000\, m\, P}{V},$$

where $c_{Zn}$ is the zinc concentration (mg/L), $m$ is the mass of the metal (mg), $P$ is its purity (dimensionless) and $V$ is the volume of the final solution (mL). The factor 1000 converts millilitres to litres.

| Input | | Type | PDF | Parameters |
|---|---|---|---|---|
| Purity $P$ | | B | Uniform | min 0.9985, max 1.0005 |
| Mass $m$ | | B | Gaussian | mean 300.75 mg, SD 0.009 mg |
| Volume $V$ | filling | B | Triangular | min 149.6 mL, max 150.4 mL, mean 150 mL |
| | repeatability | A | Rectangular | range 0.09 mL, mean 0 mL |
| | temperature | B | Normal | mean 0 mL, 0.1 mL at 95 % confidence |

The volume is therefore modelled as the sum of three components,

$$V = V_{\text{fill}} + \delta V_{\text{rep}} + \delta V_{\text{temp}}.$$

Required: the uncertainty budget (mean, standard uncertainties, sensitivity coefficients, combined standard uncertainty) and the expanded uncertainty at a 95 % confidence level.

## 2 Standard uncertainty of the assumed distributions

The standard uncertainty of an input is the standard deviation of its probability density function $g(x)$:

$$u^2 = \int_{-\infty}^{\infty} (x-\mu)^2\, g(x)\, dx.$$

### Rectangular (uniform) distribution of half-width $a$

With $g(x) = \dfrac{1}{2a}$ on $[\mu - a, \mu + a]$, substituting $t = x - \mu$:

$$u^2 = \int_{-a}^{a} t^2 \frac{1}{2a}\, dt = \frac{1}{2a}\left[\frac{t^3}{3}\right]_{-a}^{a} = \frac{a^2}{3}, \qquad u = \frac{a}{\sqrt{3}}.$$

### Symmetric triangular distribution of half-width $a$

With $g(t) = \dfrac{a - |t|}{a^2}$ on $[-a, a]$:

$$u^2 = 2\int_0^a t^2\,\frac{a-t}{a^2}\,dt = \frac{2}{a^2}\left[\frac{at^3}{3} - \frac{t^4}{4}\right]_0^a = \frac{2}{a^2}\cdot\frac{a^4}{12} = \frac{a^2}{6}, \qquad u = \frac{a}{\sqrt{6}}.$$

### Normal distribution

A standard deviation is used directly. If an expanded uncertainty $U$ is quoted at 95 % confidence for a normal distribution, the coverage factor is $k = 1.96$ and $u = U/1.96$.

## 3 Step 1: Best estimates of the inputs

$$\begin{aligned} P_0 &= \frac{P_{\min} + P_{\max}}{2} = \frac{0.9985 + 1.0005}{2} = 0.9995 \quad \text{(centre of the uniform PDF)},\\ m_0 &= 300.75 \text{ mg},\\ V_0 &= V_{\text{fill}} + \delta V_{\text{rep}} + \delta V_{\text{temp}} = 150 + 0 + 0 = 150 \text{ mL}. \end{aligned}$$

## 4 Step 2: Best estimate of the concentration

$$c_{Zn} = \frac{1000 \times m_0 \times P_0}{V_0} = \frac{1000 \times 300.75 \times 0.9995}{150} = \frac{300\,599.625}{150} = 2003.9975 \text{ mg/L}.$$

## 5 Step 3: Standard uncertainties of the inputs

### Purity $P$ (uniform distribution, Type B)

Half-width:

$$a_P = \frac{P_{\max} - P_{\min}}{2} = \frac{1.0005 - 0.9985}{2} = 0.0010,$$

$$u(P) = \frac{a_P}{\sqrt{3}} = \frac{0.0010}{1.73205} = 5.7735 \times 10^{-4}.$$

### Mass $m$ (Gaussian distribution, Type B)

The standard deviation is given directly:

$$u(m) = 0.009 \text{ mg}.$$

### Volume due to filling (triangular distribution, Type B)

Half-width:

$$a_{\text{fill}} = \frac{150.4 - 149.6}{2} = 0.4 \text{ mL},$$

$$u(V_{\text{fill}}) = \frac{a_{\text{fill}}}{\sqrt{6}} = \frac{0.4}{2.44949} = 0.16330 \text{ mL}.$$

### Volume due to repeatability (rectangular distribution, Type A)

The range of 0.09 mL is taken as the full width of the rectangular distribution, so the half-width is

$$a_{\text{rep}} = \frac{0.09}{2} = 0.045 \text{ mL},$$

$$u(\delta V_{\text{rep}}) = \frac{a_{\text{rep}}}{\sqrt{3}} = \frac{0.045}{1.73205} = 0.025981 \text{ mL}.$$

### Volume due to temperature (normal distribution, Type B)

The uncertainty of 0.1 mL is quoted at 95 % confidence for a normal distribution, so $k = 1.96$:

$$u(\delta V_{\text{temp}}) = \frac{0.1}{1.96} = 0.051020 \text{ mL}.$$

## 6 Step 4: Combined standard uncertainty of the volume

The three volume components are added, so each has a sensitivity coefficient $\partial V/\partial V_k = 1$, and

$$\begin{aligned} u^2(V) &= u^2(V_{\text{fill}}) + u^2(\delta V_{\text{rep}}) + u^2(\delta V_{\text{temp}}) \\ &= (0.16330)^2 + (0.025981)^2 + (0.051020)^2 \\ &= 0.026667 + 0.000675 + 0.002603 \\ &= 0.029945 \text{ mL}^2, \end{aligned}$$

$$u(V) = \sqrt{0.029945} = 0.17305 \text{ mL}.$$

## 7 Step 5: Sensitivity coefficients

The sensitivity coefficients are the partial derivatives of $c_{Zn} = 1000\, m\, P/V$, evaluated at the best estimates.

### With respect to mass

$$c_m = \frac{\partial c_{Zn}}{\partial m} = \frac{1000\, P}{V} = \frac{1000 \times 0.9995}{150} = \frac{999.5}{150} = 6.6633 \text{ L}^{-1}.$$

### With respect to purity

$$c_P = \frac{\partial c_{Zn}}{\partial P} = \frac{1000\, m}{V} = \frac{1000 \times 300.75}{150} = \frac{300\,750}{150} = 2005.0 \text{ mg/L}.$$

### With respect to volume

$$c_V = \frac{\partial c_{Zn}}{\partial V} = -\frac{1000\, m\, P}{V^2} = -\frac{c_{Zn}}{V} = -\frac{2003.9975}{150} = -13.360 \text{ mg L}^{-1}\,\text{mL}^{-1}.$$

Because $V$ is the sum of its three components, the same coefficient $c_V$ applies to $V_{\text{fill}}$, $\delta V_{\text{rep}}$ and $\delta V_{\text{temp}}$.

## 8 Step 6: Combined standard uncertainty

For uncorrelated inputs, the law of propagation of uncertainty gives

$$u_c^2(c_{Zn}) = c_P^2\,u^2(P) + c_m^2\,u^2(m) + c_V^2\,u^2(V).$$

### Individual contributions

$$\begin{aligned}
|c_P|\,u(P) &= 2005.0 \times 5.7735 \times 10^{-4} = 1.1576 \text{ mg/L},\\
|c_m|\,u(m) &= 6.6633 \times 0.009 = 0.0600 \text{ mg/L},\\
|c_V|\,u(V_{\text{fill}}) &= 13.360 \times 0.16330 = 2.1817 \text{ mg/L},\\
|c_V|\,u(\delta V_{\text{rep}}) &= 13.360 \times 0.025981 = 0.3471 \text{ mg/L},\\
|c_V|\,u(\delta V_{\text{temp}}) &= 13.360 \times 0.051020 = 0.6816 \text{ mg/L},\\
|c_V|\,u(V) &= 13.360 \times 0.17305 = 2.3119 \text{ mg/L}.
\end{aligned}$$

### Sum of squares

$$\begin{aligned}
u_c^2(c_{Zn}) &= (1.1576)^2 + (0.0600)^2 + (2.3119)^2\\
&= 1.3400 + 0.0036 + 5.3448\\
&= 6.6884 \text{ (mg/L)}^2,
\end{aligned}$$

$$u_c(c_{Zn}) = \sqrt{6.6884} = 2.5862 \text{ mg/L}.$$

### Check using relative uncertainties

For a model of the form $c = 1000\,m\,P/V$ (a product and quotient), the relative uncertainties add in quadrature:

$$\left(\frac{u_c(c_{Zn})}{c_{Zn}}\right)^2 = \left(\frac{u(m)}{m}\right)^2 + \left(\frac{u(P)}{P}\right)^2 + \left(\frac{u(V)}{V}\right)^2.$$

$$\begin{aligned}
\frac{u(m)}{m} &= \frac{0.009}{300.75} = 2.9925 \times 10^{-5},\\
\frac{u(P)}{P} &= \frac{5.7735 \times 10^{-4}}{0.9995} = 5.7764 \times 10^{-4},\\
\frac{u(V)}{V} &= \frac{0.17305}{150} = 1.1536 \times 10^{-3},
\end{aligned}$$

$$\begin{aligned}
\frac{u_c(c_{Zn})}{c_{Zn}} &= \sqrt{(2.9925 \times 10^{-5})^2 + (5.7764 \times 10^{-4})^2 + (1.1536 \times 10^{-3})^2}\\
&= 1.2905 \times 10^{-3} \quad (0.129\,\%),
\end{aligned}$$

$$u_c(c_{Zn}) = 1.2905 \times 10^{-3} \times 2003.9975 = 2.5862 \text{ mg/L},$$

which agrees with the value above.

## 9 Step 7: Effective degrees of freedom

None of the inputs is given with a finite number of degrees of freedom: the repeatability is specified directly as a rectangular distribution, without the number of observations from which it was obtained. All inputs are therefore treated as having $\nu_i = \infty$, so

$$\nu_{\text{eff}} = \frac{u_c^4}{\displaystyle\sum_i \frac{(c_i u(x_i))^4}{\nu_i}} = \infty.$$

## 10 Step 8: Coverage factor and expanded uncertainty

For $\nu_{\text{eff}} = \infty$ and a 95 % coverage probability, the coverage factor is that of the normal distribution, $k = 1.96$:

$$U = k\, u_c(c_{Zn}) = 1.96 \times 2.5862 = 5.069 \text{ mg/L} \approx 5.1 \text{ mg/L}.$$

With the conventional coverage factor $k = 2$:

$$U = 2 \times 2.5862 = 5.172 \text{ mg/L}.$$

The 95 % coverage interval is

$$[\, c_{Zn} - U,\ c_{Zn} + U \,] = [\, 2003.998 - 5.069,\ 2003.998 + 5.069 \,] = [\, 1998.929,\ 2009.066 \,] \text{ mg/L}.$$

## 11 Uncertainty budget

| Source | Estimate | Type | PDF | $u(x_i)$ | $c_i$ | $\lvert c_i \rvert u(x_i)$ (mg/L) | $(c_i u(x_i))^2$ $(\text{mg/L})^2$ | % |
|---|---|---|---|---|---|---|---|---|
| Purity $P$ | 0.9995 | B | Uniform | $5.7735 \times 10^{-4}$ | 2005.0 | 1.1576 | 1.3400 | 20.03 |
| Mass $m$ | 300.75 mg | B | Gaussian | 0.009 mg | 6.6633 | 0.0600 | 0.0036 | 0.05 |
| $V$, filling | 150 mL | B | Triangular | 0.16330 mL | −13.360 | 2.1817 | 4.7597 | 71.16 |
| $V$, repeatability | 0 | A | Rectangular | 0.025981 mL | −13.360 | 0.3471 | 0.1205 | 1.80 |
| $V$, temperature | 0 | B | Normal | 0.051020 mL | −13.360 | 0.6816 | 0.4646 | 6.95 |
| Sum | | | | | | | 6.6884 | 100 |

$c_{Zn} = 2003.998$ mg/L, $u_c = 2.5862$ mg/L (0.129 %), $\nu_{\text{eff}} = \infty$, $k = 1.96$
$U = 5.07$ mg/L, 95 % interval [1998.929, 2009.066] mg/L

The volume-filling tolerance dominates the budget (71.2 % of the variance), followed by the purity (20.0 %) and the temperature effect on the volume (7.0 %). The volume repeatability contributes 1.8 % and the mass is negligible (0.05 %).

## 12 Result

The zinc concentration is

$$\boxed{c_{Zn} = (2004.0 \pm 5.1) \text{ mg/L}}$$

where the number following ± is the expanded uncertainty $U = k\, u_c$, with combined standard uncertainty $u_c = 2.6$ mg/L (0.13 % relative) and coverage factor $k = 1.96$, corresponding to a coverage probability of approximately 95 %.